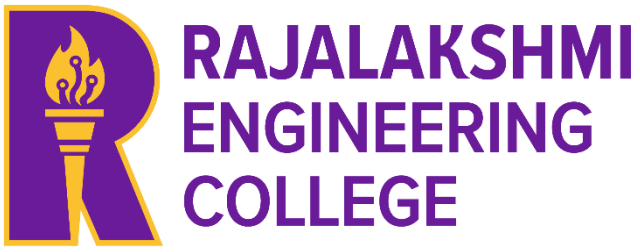
RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

Laboratory Record Note Book

Name :	:	PRAKASH A
Year / Branch / Section	:	II / BE Computer Science and Design / -
University Register No.	:	2116251701041
College Roll No. :	:	251701041
Semester:	:	III
Academic Year :	:	2026 - 2027

COMPETITIVE PROGRAMMING

EXPERIMENT 20

1-Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

Competitive Programming

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 11:34 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:36 PM
Time taken	1 min 28 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n, i, j;

    scanf("%d", &n);

    int arr[n];

    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(arr[i] == arr[j])
            {
                printf("%d\n", arr[i]);
                return 0;
            }
        }
    }

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

EXPERIMENT 21

2-Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

Competitive Programming

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 11:28 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:39 PM
Time taken	11 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int T;

    scanf("%d", &T);

    while(T--)
    {
        int n1, n2, i, j;

        scanf("%d", &n1);

        int *a = malloc(n1 * sizeof(int));

        for(i = 0; i < n1; i++)
        {
            scanf("%d", &a[i]);
        }

        scanf("%d", &n2);

        int *b = malloc(n2 * sizeof(int));

        for(i = 0; i < n2; i++)
        {
            scanf("%d", &b[i]);
        }

        i = 0;
        j = 0;

        int first = 1;

        while(i < n1 && j < n2)
        {
            if(a[i] == b[j])
            {
                if(i == 0 || a[i] != a[i - 1])
                {
                    if(!first)
                    {
                        printf(" ");
                    }

                    printf("%d", a[i]);
                    first = 0;
                }

                i++;
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

EXPERIMENT 22

3-Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Complexity

Competitive Programming

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 11:28 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:39 PM
Time taken	11 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int T;

    scanf("%d", &T);

    while(T--)
    {
        int n1, n2, i, j;

        scanf("%d", &n1);

        int *a = malloc(n1 * sizeof(int));

        for(i = 0; i < n1; i++)
        {
            scanf("%d", &a[i]);
        }

        scanf("%d", &n2);

        int *b = malloc(n2 * sizeof(int));

        for(i = 0; i < n2; i++)
        {
            scanf("%d", &b[i]);
        }

        i = 0;
        j = 0;

        int first = 1;

        while(i < n1 && j < n2)
        {
            if(a[i] == b[j])
            {
                if(i == 0 || a[i] != a[i - 1])
                {
                    if(!first)
                    {
                        printf(" ");
                    }

                    printf("%d", a[i]);
                    first = 0;
                }

                i++;
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

EXPERIMENT 23

4-Print Intersection of 2 sorted arrays-O(m+n)Time Complexity,O(1) Space Complexity

Competitive Programming

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 11:46 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:47 PM
Time taken	54 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int T;
    scanf("%d", &T);

    while(T--)
    {
        int n1, n2, i, j;

        scanf("%d", &n1);
        int a[n1];

        for(i = 0; i < n1; i++)
        {
            scanf("%d", &a[i]);
        }

        scanf("%d", &n2);
        int b[n2];

        for(i = 0; i < n2; i++)
        {
            scanf("%d", &b[i]);
        }

        i = 0;
        j = 0;

        while(i < n1 && j < n2)
        {
            if(a[i] == b[j])
            {
                if(i == 0 || a[i] != a[i - 1])
                {
                    printf("%d ", a[i]);
                }

                i++;
                j++;
            }
            else if(a[i] < b[j])
            {
                i++;
            }
            else
            {
                j++;
            }
        }

```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

5-Pair with Difference-O(n^2)Time Complexity,O(1) Space Complexity

Competitive Programming

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 11:21 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:21 PM
Time taken	45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n, a[100], k, i, j;
    scanf("%d", &n);

    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    scanf("%d", &k);

    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(a[j] - a[i] == k)
            {
                printf("1");
                return 0;
            }
        }
    }

    printf("0");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

EXPERIMENT 25

6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

Competitive Programming

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 11:44 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:45 PM
Time taken	51 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n, i, j, k;

    scanf("%d", &n);

    int a[n];

    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    scanf("%d", &k);

    i = 0;
    j = 1;

    while(i < n && j < n)
    {
        if(i != j && a[j] - a[i] == k)
        {
            printf("1");
            return 0;
        }
        else if(a[j] - a[i] < k)
        {
            j++;
        }
        else
        {
            i++;
        }

        if(i == j)
        {
            j++;
        }
    }

    printf("0");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

----END----